

The Black-Scholes Model

Because of its importance as a foundation of option pricing theory, as well as its widespread use by traders, it will be worthwhile to take a closer look at the Black-Scholes model.¹ The discussion in this chapter is not meant to be a rigorous or detailed derivation of the model, which is better suited to a university textbook or to a class in financial engineering. Rather, we hope to present a more intuitive discussion of the workings of the model, as well as some observations on the values generated by the model.

Initially, rather than calculating the theoretical value of an option, Black and Scholes tried to answer this question: if the stock price moves randomly over time, but in a manner that is consistent with a constant interest rate and volatility, what must be the option price after each moment in time such that an option position that is correctly hedged will just break even? The answer to this question resulted in rather intimidating-looking equation

$$rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + \frac{\partial C}{\partial t} = rC$$

Although this equation might look mysterious to many readers, it is just a mathematician's way of expressing how changes in one set of variables—stock price S and time t —affect the value of something else, a call C . To determine the exact effect caused by changes in the variables, one must solve the equation.

Note that we did not refer to the volatility σ and interest rate r as variables. In the Black-Scholes equation, only the stock price and time are changing. As inputs into the model, the volatility and interest rate will affect the value of the

¹The Black-Scholes model is sometimes referred to as the *Black-Scholes-Merton model* because Robert Merton, originally associated with the Massachusetts Institute of Technology, contributed significantly to the theory of option pricing. Merton and Scholes were jointly awarded the Nobel Prize in Economics in 1997 for their work on option pricing. Fischer Black, sadly, died in 1995.

option. But once they have been chosen, they are assumed to remain constant over the life of the option. This is consistent with the dynamic hedging examples in Chapter 8. Over the life of an option, we assumed that only the underlying price and time were changing. Everything else remained constant.

We have already encountered several of the components of the Black-Scholes equation in slightly different form. The terms

$$\frac{\partial C}{\partial S}, \quad \frac{\partial^2 C}{\partial S^2}, \quad \text{and} \quad \frac{\partial C}{\partial t}$$

are the more formal mathematical notation for the option's delta (Δ), gamma (Γ), and theta (Θ). The Black-Scholes equation states that changes in an option's value depend on the sensitivity of the option to changes in the stock price (the delta), the sensitivity of the option's delta to changes in the stock price (the gamma), and the sensitivity of the option to the passage of time (the theta).

Of course, the equation also includes volatility and interest-rate components. The interest-rate component plays two roles. First, because the Black-Scholes model values options from the forward price, the interest rate takes us from the spot price to the forward price (assuming that the stock pays no dividend). This spot-to-forward relationship appears in the equation as

$$rS$$

Second, the Black-Scholes equation initially gives us the expected value of the option as time passes. If we want to determine the option's theoretical value, we must discount the expected value backwards to get its present value. This expected-value-to-present-value relationship appears in the equation as

$$rC$$

Finally, there is a volatility component. The rate at which the delta changes depends not only on the gamma but on the speed at which the stock price is changing. The speed is expressed as a volatility or standard deviation σ . The volatility component and its effect on the gamma appear in the Black-Scholes equation as

$$\frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} = \frac{1}{2} \sigma^2 S^2 \Gamma$$

We will not go into the formal derivation of the Black-Scholes equation in this text because it can be mathematically complex. But we might note that there is some similarity between the Black-Scholes equation and the method used in Chapter 7 to estimate the change in an option's value as the underlying price changes from S_1 to S_2 . To approximate this change, we used the average delta over the price range

$$(S_1 - S_2) \times \Delta + (S_1 - S_2)^2 \times \Gamma / 2 = (S_1 - S_2) \times \Delta + \frac{1}{2} (S_1 - S_2)^2 \times \Gamma$$

Recalling that

$$\frac{\partial C}{\partial S} \quad \text{and} \quad \frac{\partial^2 C}{\partial S^2}$$

represent the delta and gamma, we can see that there is a similarity between this relationship and the first two terms of the Black-Scholes equation.

$$(S_1 - S_2) \times \Delta + \frac{1}{2}(S_1 - S_2)^2 \times \Gamma \\ rS \frac{\partial C}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 C}{\partial S^2}$$

The primary differences are the interest-rate component attached to S (the stock price must move from spot to forward) and the volatility component attached to the gamma. Although we assumed a discrete price change from S_1 to S_2 , the Black-Scholes equation assumes an infinitesimally small, or instantaneous, price change.

This is, admittedly, a very simplistic attempt to explain the roles played by the various components in the Black-Scholes equation. However, even for someone who fully understands the model, being able to write out the equation does not necessarily yield a value. The real goal is to solve the equation so that it is possible to calculate the exact value of an option.

The solution to the Black-Scholes equation yields the well-known *Black-Scholes model*: if

C = theoretical value of a European call

S = the price of a non-dividend-paying stock

X = exercise price

t = time to expiration, in years

σ = annual standard deviation (volatility) of the stock price, in percent

r = annual interest rate

$\ln$ = the natural logarithm

e = the exponential function

N = the cumulative normal distribution function

then

$$C = SN(d_1) - Xe^{-rt}N(d_2)$$

where

$$d_1 = \frac{\ln(S/X) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}}$$

and

$$d_2 = \frac{\ln(S/X) + [r + (\sigma^2/2)]t}{\sigma\sqrt{t}} = d_1 - \sigma\sqrt{t}$$

It may not be immediately apparent what the values in the Black-Scholes model represent, but one starting point is put-call parity, discussed in Chapter 15

$$C - P = \frac{F - X}{1 + r \times t}$$

If the underlying contract is a non-dividend-paying stock, the forward price is

$$F = S \times (1 + r \times t)$$

Substituting this into the put-call parity relationship gives us

$$C - P = \frac{S(1 + r \times t) - X}{1 + r \times t} = S - \frac{X}{1 + r \times t}$$

In our examples thus far, we have used simple interest. If, instead, we use continuous interest, rather than dividing by $1 + r \times t$, we can multiply by e^{-rt} . This gives us

$$C - P = S - Xe^{-rt}$$

Because a put can never be worth less than 0, we know from Chapter 16 that the lower arbitrage boundary for a European call option on stock is the greater of either 0 or

$$S - Xe^{-rt}$$

This expression looks similar to the Black-Scholes value for a call option, but without the terms $N(d_1)$ and $N(d_2)$ attached to S and Xe^{-rt} , respectively. What do $N(d_1)$ and $N(d_2)$ represent?

In Chapter 5, we proposed a very simple method for evaluating options by considering a series of underlying prices at expiration and assigning probabilities to each of those prices. Using this approach, the expected value for a call option is the sum of the intrinsic values multiplied by the probability associated with each underlying price

$$\sum_{i=1}^n p_i \max(S_i - X, 0)$$

To determine the option's intrinsic value, we combined the underlying price and exercise price into one expression ($S_i - X$).

The Black-Scholes model takes a slightly different approach by separating the underlying price and exercise price into two distinct components and then asking two questions:

1. If held to expiration, what is the average value of all the stock above the exercise price?
2. If held to expiration, what is the likelihood that the owner of an option will end up paying the exercise price?

If we can answer these questions, the difference between the average value of the stock above the exercise price and the likelihood of paying the exercise price should equal the option's expected value.

To help explain the approach taken by Black and Scholes, let's consider a discrete distribution of stock prices at expiration, but one that more closely resembles a lognormal distribution with an extended right tail. Such a distribution, resulting from a total of 153 occurrences, is shown in Figure 18-1. Using this distribution, how might we evaluate a call option with an exercise price of $12\frac{1}{2}$?

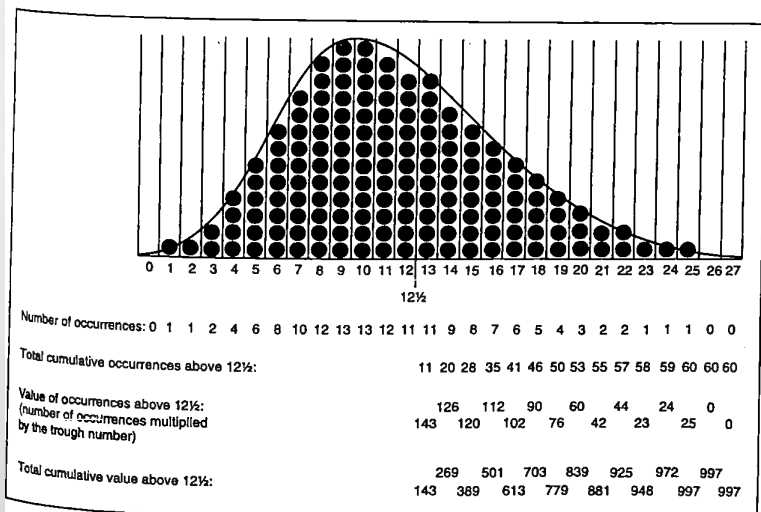
First, we must determine the value of all stock above $12\frac{1}{2}$, that is, the value resulting from all occurrences that fall into troughs 13 through 27. The number of occurrences and the value of the occurrences in each trough are as follows:

Trough	Number of Occurrences	Stock Value
13	11	143
14	9	126
15	8	120
16	7	113
17	6	102
18	5	90
19	4	76
20	3	60
21	2	42
22	2	44
23	1	23
24	1	24
25	1	25
26	0	0
27	0	0
Total	60	987

The average value of all stock above the exercise price of $12\frac{1}{2}$ is the total value, 987, divided by the total number of occurrences, 153

$$987/153 = 6.45$$

Figure 18-1



Next, we need to determine the likelihood that we will pay the exercise price of $12\frac{1}{2}$. There are 60 occurrences where the option is in the money (the stock price is above $12\frac{1}{2}$), but there are a total of 153 occurrences. The likelihood that we will pay the exercise price is

$$60/153 = 0.392$$

The average payout resulting from exercise of the option $0.392 \times 12\frac{1}{2} = 4.90$.

In the Black-Scholes model, the average value of all stock above the exercise price is given by $Se^rN(d_1)$, where Se^r is the forward price of the stock. The average amount we will have to pay is given by $KN(d_2)$. The expected value for a call option is the difference between these two numbers

$$Se^rN(d_1) - KN(d_2) = 6.45 - 4.90 = 1.55$$

These terms are slightly different from the terms that appear in the model, $SN(d_1)$ and $Xe^{-r}N(d_2)$, but we will show shortly how $Se^rN(d_1)$ becomes $SN(d_1)$ and how $KN(d_2)$ becomes $Xe^{-r}N(d_2)$.

We can confirm that 1.55 is the correct value (with slight rounding error) by returning to our original approach of adding up the intrinsic values multiplied by their probabilities (the number of occurrences divided by 153).

Trough	Intrinsic Value of the $12\frac{1}{2}$ Call	Number of Occurrences	Probability	Option Value
13	0.5	11	0.0719	0.0359
14	1.5	9	0.0588	0.0882
15	2.5	8	0.0523	0.1307
16	3.5	7	0.0458	0.1601
17	4.5	6	0.0392	0.1765
18	5.5	5	0.0327	0.1797
19	6.5	4	0.0261	0.1699
20	7.5	3	0.0196	0.1471
21	8.5	2	0.0131	0.1111
22	9.5	2	0.0131	0.1242
23	10.5	1	0.0065	0.0686
24	11.5	1	0.0065	0.0752
25	12.5	1	0.0065	0.0817
26	13.5	0	0	0
27	14.5	0	0	0
Total option expected value:			1.5489	

This is essentially the approach taken by the Black and Scholes. The primary difference is that the Black-Scholes model, rather than using discrete outcomes as we did, assumes a continuous lognormal distribution.

$n(x)$ and $N(x)$

Before continuing, it will be useful to define two important probability functions— $n(x)$ and $N(x)$. In this chapter and in previous discussions of volatility, we have often referred to the concept of a bell-shaped, or normal, distribution. Depending on the mean and standard deviation, there can be many different normal distributions, but $n(x)$, the *standard normal distribution*, is perhaps the most common. It has a mean of 0 and a standard deviation of 1. The standard normal distribution, shown in Figure 18-2, also has one very useful characteristic: the total area under the curve adds up to exactly 1. That is, the curve represents 100 percent of all occurrences that form a true normal distribution.

Although the standard normal distribution takes in 100 percent of all occurrences, we may want to know what percent of the occurrences fall within a specific portion of the standard normal distribution. This is given by $N(x)$, the *standard cumulative normal distribution function*. If x is some number of standard deviations, $N(x)$ returns the probability of getting an occurrence less than x by calculating the area under the standard normal distribution curve between the values of $-\infty$ and x , as shown in Figure 18-3. That is, $N(x)$ tells us what percentage of all possible occurrences fall between $-\infty$ and x . Obviously, $N(+\infty)$ must be 1.00 because 100 percent of all occurrences must fall between $-\infty$ and $+\infty$. And $N(-\infty)$ must be 0 because there can be no occurrences to the left of $-\infty$. Because the normal distribution curve is symmetrical, with 50 percent of the occurrences falling to the left of 0 and 50 percent falling to the right, $N(0)$ must equal 0.50. It also follows that the area under the curve between $-\infty$ and x must be equal to the area under the curve between $-x$ and $+\infty$, resulting in this useful relationship

$$N(x) = 1 - N(-x)$$

Figure 18-2 $n(x)$ —the standard normal distribution curve with mean = 0 and standard deviation = 1.

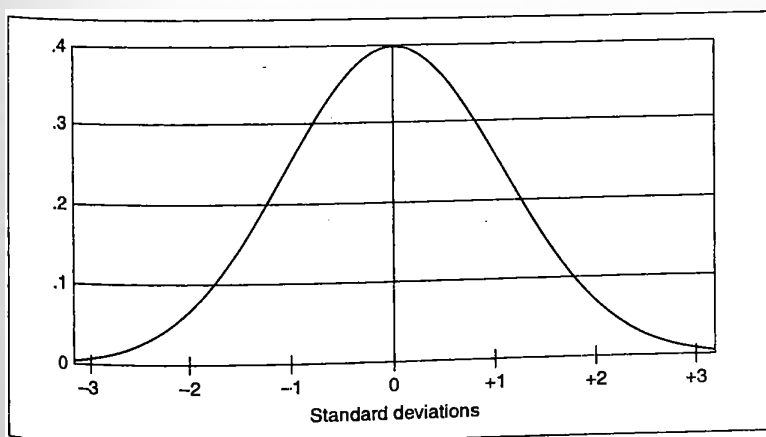
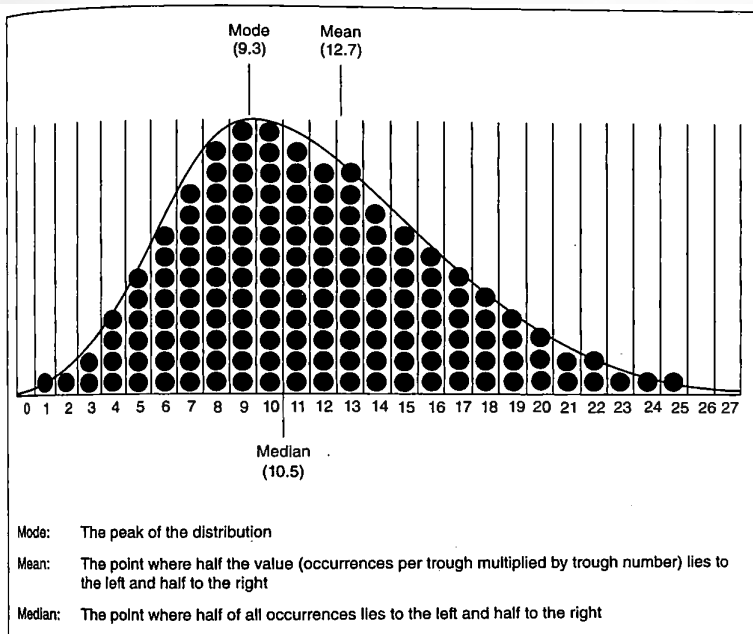


Figure 18-4



If $S > X$, this value is positive, and the call is in the money; if $S < X$, the value is negative, and the call is out of the money.

Next, because options are valued off the forward price and the forward price is a function of interest rates, we must adjust this relationship by the interest component over the life of the option τ . This gives us²

$$\ln\left(\frac{S}{X}\right) + \tau$$

The number of standard deviations associated with an occurrence depends on how far the occurrence is from the mean of the distribution. In a normal distribution, the mean, like the mode, is located in the exact center of the distribution. But in Figure 18-4, which approximates a lognormal distribution, with its elongated right tail, we can see that the mean must be somewhere to the right of the mode. How far to the right? This depends on the standard deviation of the lognormal distribution. The higher the standard deviation, the longer the right tail, and consequently, the further to the right we must shift the mean. Mathematically, the shift is equal to $\sigma^2 t / 2$. Adding this adjustment gives us

$$\ln\left(\frac{S}{X}\right) + \tau + \frac{\sigma^2 t}{2}$$

² We could in fact drop τ and at the same time replace S with its forward price S^f . The values are the same: $\ln(S/X) + \tau = \ln(S^f/X)$.

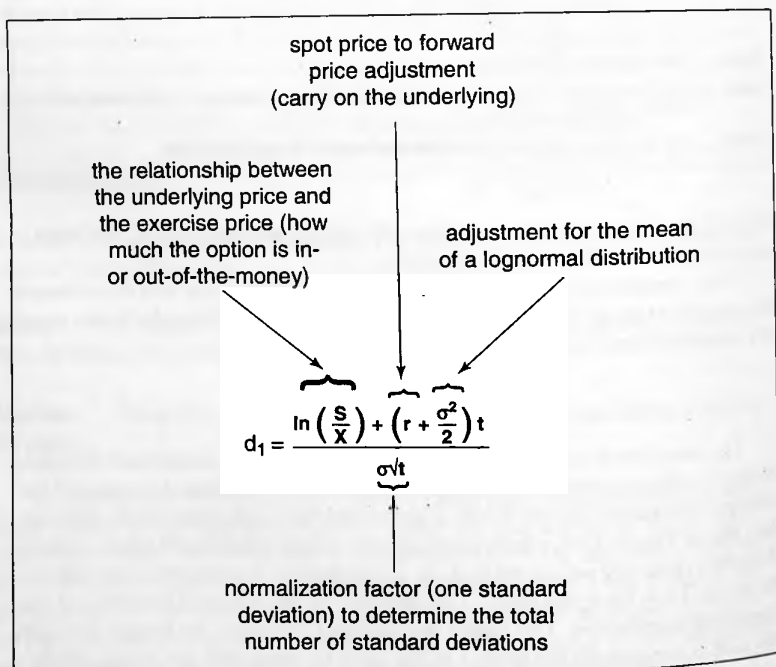
Combining the interest-rate and volatility components gives us the numerator for d_1

$$\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)t$$

Finally, we must convert this value to some number of standard deviations. If we know the value of one standard deviation, we can divide by this value to determine the total number of standard deviations. In fact, we know that over any time period t , one standard deviation is equal to $\sigma\sqrt{t}$. If we divide by this value, the result, d_1 , tells us, in standard deviations, how far the exercise price is from the mean when adjusted for a lognormal distribution

In the equation shown in Figure 18-5, the calculation of d_1 may seem somewhat complicated, but it is really just a collection of adjustments to the exercise price and underlying price that enable us to use a cumulative normal distribution function to calculate probabilities.

Figure 18-5



Once we have determined the value of d_1 , multiplying the forward price of the stock by $N(d_1)$ gives us the average value of all stock above the exercise price at expiration.

Having calculated the average value of all stock above the exercise price, we still need to determine the likelihood that the option will be exercised. To do this, we need the median of the distribution, the point that exactly bisects the total number of occurrences. In Figure 18-4, we can see that the median in a lognormal distribution falls somewhere to the left of the mean. How far to the left? In fact, the median falls to the left by $\sigma\sqrt{t}$

$$d_2 = d_1 - \sigma\sqrt{t}$$

The value $N(d_2)$ uses the median to calculate the probability of the option being in the money at expiration and therefore being exercised. Multiplying this probability by the exercise price gives us the average amount we will pay at expiration if we own the option

$$XN(d_2)$$

Taking the average value of the stock we will receive at expiration and subtracting the average amount we will pay at expiration gives us the expected value for the call

$$Se^rN(d_1) - XN(d_2)$$

There is still one final step in calculating the theoretical value of a call option, and this step explains how the terms $S^rN(d_1)$ and $XN(d_2)$ become $SN(d_1)$ and $Xe^{-r}N(d_2)$, which is the way they appear in the Black-Scholes model. The expression $Se^rN(d_1) - XN(d_2)$ represents the expected value of the option at expiration. If we must pay for the option today, the theoretical value is the present value of the expected value. Multiplying the expected value by e^{-r} yields the familiar form of the Black-Scholes model

$$C = [Se^rN(d_1) - XN(d_2)]e^{-r} = SN(d_1) - Xe^{-r}N(d_2)$$

In the original Black-Scholes model, the underlying contract was assumed to be a non-dividend-paying stock. However, since its introduction, the model has been extended to evaluate options on other types of underlying instruments. This is most commonly done by including an adjustment factor b that varies depending on the type of underlying instrument and the settlement procedure for the options. If r is the domestic interest rate and r_f is the foreign interest rate, then

- $b = r$ The original Black-Scholes model for options on a non-dividend-paying stock
- $b = 0$ The Black model for options on futures
- $b = r - r_f$ The Garman-Kohlhagen model for options on foreign currencies

The complete Black-Scholes model, with variations and sensitivities, is given in Figure 18-6.

Figure 18-6 The Black-Scholes model.

If	S = the spot price or underlying price X = the exercise price t = the time to expiration, in years r = the domestic interest rate σ = the annualized volatility or standard deviation, in percent	
then the value of a European call, C , and the value of a European put, P , are given by		
	$C = S e^{(b-r)t} N(d_1) - X e^{-rt} N(d_2)$	$P = X e^{-rt} N(-d_2) - S e^{(b-r)t} N(-d_1)$
where	$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(b + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$	$d_2 = \frac{\ln\left(\frac{S}{X}\right) + \left(b - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} = d_1 - \sigma\sqrt{t}$
The common variations on the original Black-Scholes model are determined by the value of b .		
If	$b = r$: The Black-Scholes model for options on stock $b = r = 0$: The Black-Scholes model for options on futures where the options are subject to futures-type settlement $b = 0$: The Black model for options on futures where the options are subject to stock-type settlement $b = r - r_f$: The Garman-Kohlhagen model for options on foreign currencies, where r_f is the foreign interest rate	
For options on a dividend paying stock the spot price, S, must be discounted by the value of the expected dividend payments. This can be approximated by setting $b = r - q$, where q is the annual dividend yield in percent. For a more exact calculation we can deduct from S the value of each dividend payment, D, together with the interest which can be earned on that dividend payment to expiration. S is then replaced by $S - \sum D_i e^{-rt_i}$, where t_i is the time remaining from each dividend payment to expiration of the option.		

	Call		Put
Delta (Δ)	$e^{(b-r)t} N(d_1)$		$e^{(b-r)t} [N(d_1) - 1]$
Gamma (Γ)	$\frac{e^{(b-r)t} n(d_1)}{S\sigma\sqrt{t}}$	same for calls and puts	$\frac{e^{(b-r)t} n(d_1)}{S\sigma\sqrt{t}}$
Theta (Θ)*	$\frac{-Se^{(b-r)t} n(d_1)\sigma}{2\sqrt{t}} + (b-r)Se^{(b-r)t} N(d_1) - rXe^{-rt} N(d_2)$		$\frac{-Se^{(b-r)t} n(d_1)\sigma}{2\sqrt{t}} + (b-r)Se^{(b-r)t} N(d_1) + rXe^{-rt} N(-d_2)$
Vega**	$Se^{(b-r)t} n(d_1)\sqrt{t}$	same for calls and puts	$Se^{(b-r)t} n(d_1)\sqrt{t}$
Rho (ρ)**	$tXe^{-rt} N(d_2)$ if $b \neq 0$ $-tC$ if $b = 0$		$-tXe^{-rt} N(-d_2)$ if $b \neq 0$ $-tP$ if $b = 0$
Rho _f or Phi (Φ)	$-tSe^{(b-r)t} N(d_1)$		$tSe^{(b-r)t} N(-d_1)$
Elasticity (Λ)	$\Delta_C (S/C)$		$\Delta_P (S/P)$

*The theta formula gives the sensitivity of the option to the passage of one year. To express theta values in the more common form of daily decay the theta must be divided by 365.

**The vega and rho formulas give the sensitivity of the option to a one full point (100%) change in volatility (the vega) or interest rates (rho). To express vega and rho values in the more common form of a one percentage point change in volatility or interest rates the vega and rho must be divided by 100.

(continued)

Figure 18-6 (continued)

	Call		Put
Vanna	$-e^{(b-r)t} n(d_1) \frac{d_2}{\sigma}$	same for calls and puts	$-e^{(b-r)t} n(d_1) \frac{d_2}{\sigma}$
Charm	$-e^{(b-r)t} \left[n(d_1) \left(\frac{b}{\sigma\sqrt{t}} - \frac{d_2}{2t} \right) + (b-r)N(d_1) \right]$		$-e^{(b-r)t} \left[n(d_1) \left(\frac{b}{\sigma\sqrt{t}} - \frac{d_2}{2t} \right) - (b-r)N(d_1) \right]$
Speed	$-\frac{\Gamma}{S} \left(1 + \frac{d_1}{\sigma\sqrt{t}} \right)$	same for calls and puts	$-\frac{\Gamma}{S} \left(1 + \frac{d_1}{\sigma\sqrt{t}} \right)$
Color	$\Gamma \left(r - b + \frac{bd_1}{\sigma\sqrt{t}} + \frac{1-d_1d_2}{2t} \right)$	same for calls and puts	$\Gamma \left(r - b + \frac{bd_1}{\sigma\sqrt{t}} + \frac{1-d_1d_2}{2t} \right)$
Volga (Vomma)	$\text{vega} \left(\frac{d_1d_2}{\sigma} \right)$	same for calls and puts	$\text{vega} \left(\frac{d_1d_2}{\sigma} \right)$
Vega Decay	$\text{vega} \left(r - b + \frac{bd_1}{\sigma\sqrt{t}} - \frac{1-d_1d_2}{2t} \right)$	same for calls and puts	$\text{vega} \left(r - b + \frac{bd_1}{\sigma\sqrt{t}} - \frac{1-d_1d_2}{2t} \right)$
Zomma	$\Gamma \left(\frac{d_1d_2 - 1}{\sigma} \right)$	same for calls and puts	$\Gamma \left(\frac{d_1d_2 - 1}{\sigma} \right)$
A complete listing of all sensitivities and their formulas can be found in <i>The Complete Guide to Option Pricing Formula</i> by Espen Gaarder Haug, 2nd Edition, 2007, McGraw-Hill.			

A Useful Approximation

A trader might wonder whether it is possible to calculate a Black-Scholes value without using a computer. In general, the answer is no; the computations are just too complex. However, there is one type of approximation that many traders are able to make without too much difficulty.

Suppose that an option is exactly at the money ($X = S$) and that there is one year to expiration ($t = 1$). Suppose also that the interest rate is 0 ($r = 0$) and that volatility is 1 percent ($\sigma = 0.01$). This means that $\ln(S/X) = 0$ and that $\sigma\sqrt{t} = 0.01$. Calculating d_1 and d_2 , we get

$$d_1 = \frac{0.01^2/2}{0.01} = 0.005 \quad \text{and} \quad d_2 = \frac{-0.01^2/2}{0.01} = -0.005$$

If we calculate $N(d_1)$ and $N(d_2)$, we find that

$$N(d_1) = 0.501995 \quad \text{and} \quad N(d_2) = 0.498005$$

Because the interest rate is 0, the value of the call option must be

$$(S \times 0.501995) - (X \times 0.498005)$$

If $X = S$, the value of the call is

$$X \times (0.501995 - 0.498005) = X \times 0.003990$$

What does this number tell us? For a one-year European option that is exactly at the forward (i.e., the forward price is equal to the exercise price), for each percentage point of volatility, the expected value for the option is equal to the exercise price multiplied by 0.00399. If the exercise price is 100, the expected value is $0.00399 \times 100 = 0.399$ for each percentage point in volatility.

Why doesn't this value change as we increase volatility? Although the first percentage point of volatility may be worth 0.00399, perhaps the second percentage point is worth either more or less than 0.00399. But recall from Chapter 9 that the vega of an at-the-money option is relatively constant with respect to changes in volatility. Therefore, at a volatility of 20 percent, the value of a 100 call should be

$$20 \times 100 \times 0.00399 = 7.98$$

At a volatility of 35 percent, the value should be

$$35 \times 100 \times 0.00399 = 13.965$$

We also know that the theoretical value of an at-the-forward option is proportional to its exercise price. If the value of a one-year 100 call at a volatility of 20 percent is 7.98, under the same conditions, the value of an at-the-forward 50 call should be

$$20 \times 50 \times 0.00399 = 3.99$$

and the value of a 125 call should be

$$20 \times 125 \times 0.00399 = 9.975$$

We can further refine our approximation if we note that an at-the-money option is made up entirely of time value and that the time value of an option is proportional to the square root of time. If a one-year 100 call is worth 7.98 at a volatility of 20 percent, the same call with six months to expiration ($t = 0.5$) must be worth

$$7.98 \times \sqrt{0.5} = 7.98 \times 0.707 = 5.64$$

Lastly, this is an approximation for the expected value. To determine the theoretical value, we must discount by interest to get the present value. Putting everything together, for an exactly at-the-forward European option, the expected value at expiration is approximately³

$$X \times (\sigma \times 100) \times \sqrt{t} \times 0.00399$$

³To further simplify this approximation, many traders round .00399 to .004. This leads to what is sometimes referred to as the *40% rule*: the expected value of an at-the-forward option is equal to approximately 40% of one standard deviation, where one standard deviation is equal to $F \times \sigma \sqrt{t}$.

and the theoretical value is⁴

$$\frac{X \times (\sigma \times 100) \times \sqrt{t} \times 0.00399}{1 + r \times t}$$

This approximation applies to both calls and puts because under put-call parity, an exactly at-the-forward European call and put must have the same value.

For example, if volatility is 18 percent, what is the expected value of a three-month ($t = 1/4$) at-the-forward option with an exercise price of 65?

$$65 \times 0.18 \times 100 \times \sqrt{1/4} \times 0.00399 = 65 \times 18 \times 1/2 \times 0.00399 \approx 2.33$$

If interest rates are 4 percent, the option's theoretical value is approximately

$$\frac{2.33}{1 + 0.04/4} = \frac{2.33}{1.01} \approx 2.31$$

Although this is a commonly used approximation, it is only an approximation. As we increase time and volatility, the approximation will actually be slightly greater than the true Black-Scholes value. This is because the vega of an at-the-money option declines slightly as we increase volatility, and this decline is magnified with greater time to expiration. This can be seen in Figure 9-14: the vega of an at-the-money option, although relatively constant with respect to changes in volatility, does in fact decline slightly with increasing volatility. If, in our example, we raise the volatility to 40 percent and increase the time to expiration to two years, the approximation for the expected value is

$$65 \times 40 \times \sqrt{2} \times 0.00399 = 65 \times 40 \times 1.414 \times 0.00399 \approx 14.67$$

while the actual Black-Scholes expected value is 14.48.

The reader who is familiar with the characteristics of a standard normal distribution may already have recognized the significance of the value 0.00399. Referring to Figure 18-2, for a standard normal distribution with a mean of 0 and standard deviation of 1, the peak of the distribution has a value of approximately 0.399 (more exactly, 0.398942). Because a volatility of 1 percent represents 1/100 of a standard deviation, the value from the model is $0.399/100 = 0.00399$.

The Delta

In the Black-Scholes model, the delta of an option is equal to $N(d_1)$. When we defined the delta in Chapter 7, we suggested that the delta is approximately the probability that an option will finish in the money. But we now know that the true probability that an option will finish in the money is equal to $N(d_2)$. Although $N(d_1)$ and $N(d_2)$ are often very close in value, especially for short-term options, $N(d_1)$ (the delta) is always larger than $N(d_2)$.

⁴For a more exact calculation, $1 + r \times t$ can be replaced by e^{rt} .

For a call option that is at the forward, the delta will be greater than 50, even if only slightly. Because we know that

$$\text{Put delta} = \text{call delta} - 100$$

the delta of a put will be less than -50 in absolute value. This means that an at-the-forward straddle will have a positive delta. If a call and put have the same exercise price, at what forward price will the delta of the call and put be identical? This will occur when d_1 is exactly 0. A straddle will therefore be exactly delta neutral when

$$\ln\left(\frac{S}{X}\right) + [r + (\sigma^2/2)]t = 0$$

Solving, for S , we get

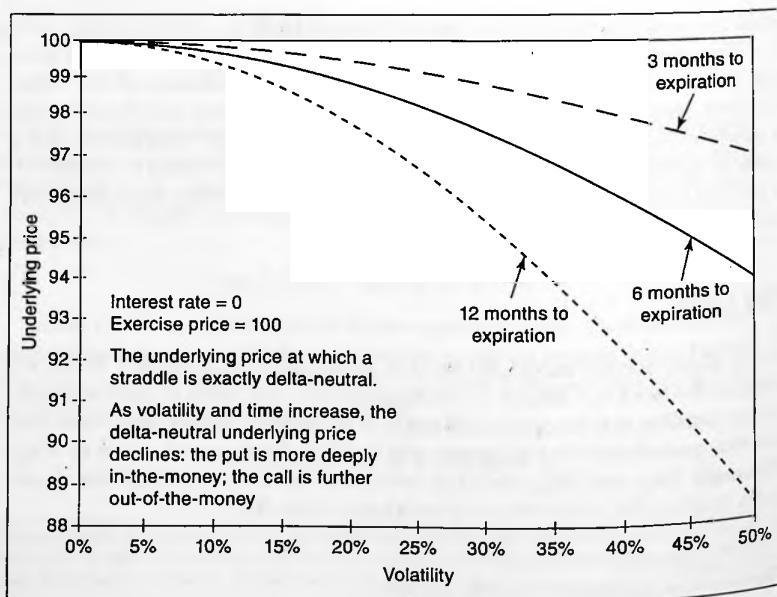
$$S = Xe^{-[r + (\sigma^2/2)]t}$$

For a straddle to be exactly delta neutral, the forward price will be less than the exercise price by a factor of

$$e^{-[r + (\sigma^2/2)]t}$$

As time or volatility increases, the forward price at which the straddle is delta neutral drops further and further below the exercise price—the call goes further out of the money, and the put goes further into the money. With a 0 interest rate, the underlying price at which a 100 straddle will be exactly delta neutral is shown in Figure 18-7. At very low volatilities, the delta-neutral price

Figure 18-7 The underlying price at which a straddle is exactly delta-neutral.



is close to 100. But, at very high volatilities and with increasing time to expiration, the delta-neutral price is well below 100.

The Theta

Of all the sensitivities derived from the Black-Scholes model, the formula for theta is probably the most complex. Depending on the underlying instrument and the option settlement procedure, the passage of time affects option values in three different ways. First, there is a decay in the option's volatility value—as time passes, the distribution of possible prices at expiration becomes more restricted. This is represented by the first term in the theta formula

$$\frac{-Se^{(b-r)t}n(d_1)\sigma}{2\sqrt{t}}$$

Second, for an underlying contract such as stock, the spot price is assumed to move toward the forward price as time passes. This is represented by the second term in the theta formula

$$(b-r)Se^{(b-r)t}N(d_1)$$

Finally, the present value of the option's expected value at expiration is changing as time passes. This appears in the formula as

$$rXe^{-rt}N(d_2)$$

We know from put-call parity that the volatility value for a call and put with identical contract specifications must be the same. The sign of the first component, the decay in volatility value, must therefore be the same for calls and puts. The other two theta components depend on the effects of interest rates and may be either positive or negative depending on the settlement procedure and whether the option is a call or a put.

The decay in volatility value is almost always more important than interest considerations and will tend to dominate the theta calculation. If interest rates are 0 or if options on futures are subject to futures-type settlement, the second and third components in the theta formula will be 0, leaving only the volatility decay component. In this case, the volatility decay component, sometimes referred to as the *driftless theta*, will be the sole factor that determines how an option's theoretical value changes as time passes.

Maximum Gamma, Theta, and Vega

In Chapter 7, we suggested that an option has its maximum gamma, theta, and vega when it is exactly at the money. But, just as we tend to assign a delta of 50 to an at-the-money option, this is only an approximation. Where does the maximum gamma, theta, and vega really occur?

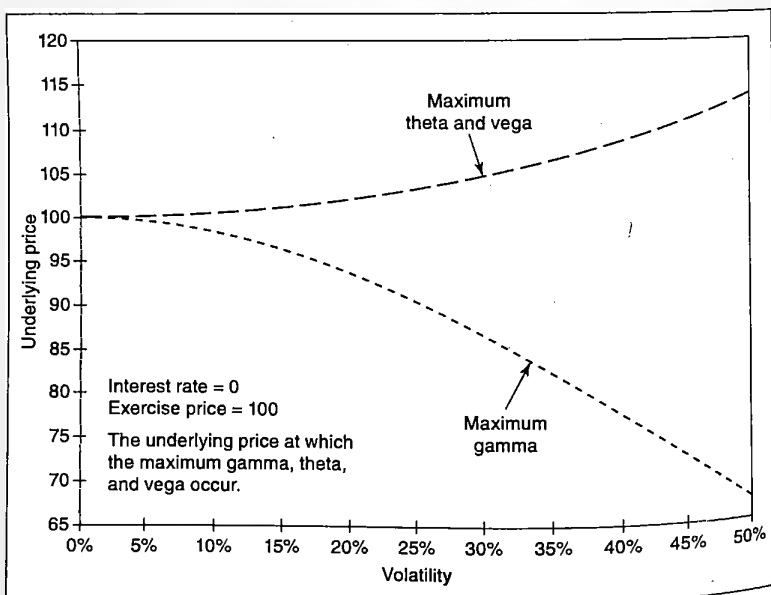
Without going into the mathematical derivation, we can summarize the critical underlying prices S as follows:

Delta of 50:	$S = Xe^{(-b-\sigma^2/2)t}$
Maximum gamma ⁵ :	$S = Xe^{(-b-3\sigma^2/2)t}$
Maximum theta ⁵ :	$S = Xe^{(b+\sigma^2/2)t}$
Maximum vega ⁵ :	$S = Xe^{(-b+\sigma^2/2)t}$

If $b = 0$, the maximum gamma and theta will occur at an underlying price that is higher than the exercise price, and the maximum vega will occur at an underlying price that is lower than the exercise price. Moreover, the maximum gamma and theta will occur at the same underlying price. This is shown in Figure 18-8 for a one-year option with an exercise price of 100. If we raise interest rates ($b > 0$), the underlying price at which the maximum gamma and vega occur will fall, and the underlying price where the maximum theta occurs will rise. This is shown in Figure 18-9.

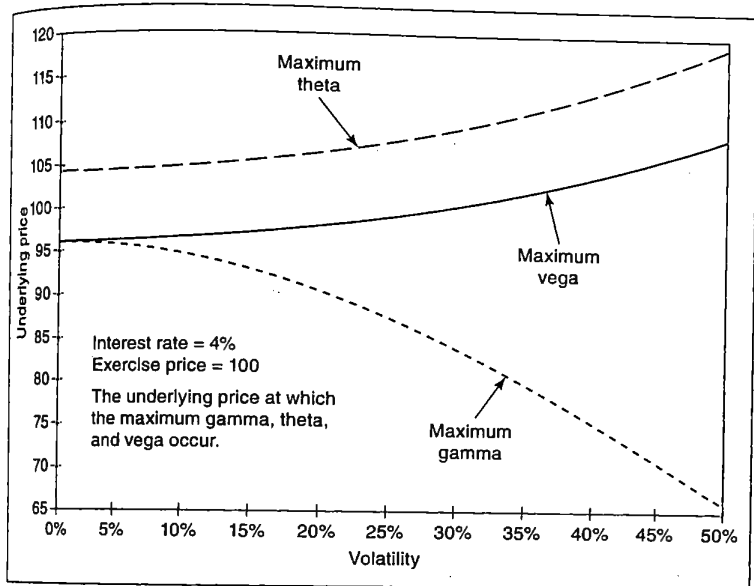
We might also consider what will happen to the vega of an option as we change time. The answer may seem obvious because we previously made the assumption that the vega always increases with time—long-term options are

Figure 18-8 At an interest rate of zero, the underlying price at which the maximum gamma, theta, and vega occur.*



*We can also relate the critical underlying prices to the higher-order risk measures. If we ignore the extremes, where the option is either very deeply in the money or very far out of the money, the maximum gamma will occur when the option's *speed* is 0. The maximum theta will occur when the option's *charm* is 0. The maximum vega will occur when the option's *vanna* is 0.

Figure 18-9 At an interest rate of 4 percent, the underlying price at which the maximum gamma, theta, and vega will occur.



more sensitive to a change in volatility than short-term options. But this is true only if the underlying price is equal to the forward price, as it is assumed to be when evaluating options on futures. If we evaluate a stock option, the forward price for stock is a function of both time and interest rates. If interest rates are greater than 0, and assuming no dividends, as we increase time, the forward price will increase, causing the option to become either more or less at the forward. Because an at-the-forward option tends to have the highest vega, changing time can cause the vega of an option to either rise or fall. This means that under some conditions, it is possible for the vega of a stock option to decline if we increase to expiration. We can see this effect in Figure 18-10.

With an underlying stock price of 100, a volatility of 20 percent, and interest rate of 0, the vega of a 100 call always increases as we increase time to expiration. But as we raise interest rates, there is some point in time at which the opposite occurs—the option's vega begins to decline as we increase time to expiration. At an interest rate of 10 percent, this occurs if there are more than 33 months remaining to expiration. At an interest rate of 20 percent, the critical time is 10 months remaining to expiration.

We can also see where these critical points are by looking at a graph of the vega decay, as shown in Figure 18-11. At an interest rate of 0, the vega decay is always positive. At an interest rate of 10 percent, the vega decay is positive with less than 33 months to expiration but negative with more than 33 months. And at an interest rate of 20 percent, the vega decay is positive with less than 10 months to expiration and negative with more than 10 months.

Figure 18-10 Vega as time and interest change.

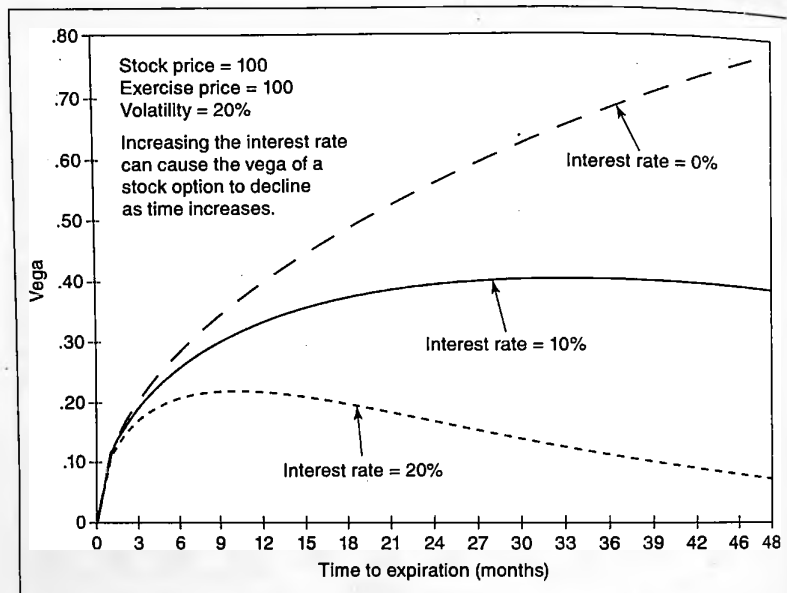


Figure 18-11 Vega decay as time and interest change.

